

Minimization of expensive functions in the presence of forbidden regions *

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Abstract

In estimating parameters for models that reflect the spatiotemporal evolution of a physical system, minimizing functions representing data error incurs a high computational cost because each evaluation requires solving a typically expensive “direct problem”. Furthermore, these functions may be undefined for parameter sets whose description via equations or inequalities is impossible. This paper addresses both challenges. To handle minimization with forbidden points, we define a general algorithm. On the other hand, to cope with the high cost of function evaluations, we propose algorithms based on minimizing surrogate models at each iteration, whose cost is affordable compared to evaluating the objective function and gradient. These techniques are applied to the problem of estimating the hydraulic parameters in the equations governing flow behavior in natural channels.

Key words: River flow modeling,

1 Introduction

We are interested in optimization problems that deal with optimizing parameters of time-evolving systems. Often, problems of this type arise from the discretization of partial differential equations with a space-time domain.

The cost of evaluating the objective function is generally high because this evaluation requires solving the underlying system. The objective function is typically a sum of squares corresponding to the available observations, and its gradient can generally be computed using automatic differentiation tools.

On the other hand, for some choices of the parameters to be estimated, the system may not be well-defined (blow-up), and it is not possible to foresee a priori in which regions this numerical event may occur. When the trial point of an iterative minimization method turns out to be a forbidden point—in the sense that the objective function is undefined—the usual practice is to step back the trial point toward the current point, trusting that the search process will encounter these anomalies infrequently.

Clearly, interpolation strategies for this backtracking are prohibitive because there is no objective function value at the trial point that can be used; furthermore, if the user decides to assign it a very large value, cubic or quadratic interpolation lands on points whose distance to the current point is negligible, thereby yielding little or no useful information.

The problems we have in mind have the following characteristics:

- (a) Expensive objective function and available expensive gradient.
- (b) Existence of forbidden points with unknown locations.

In Section 2 of this paper, we address the treatment of forbidden regions in the context of unconstrained optimization. To this end, we define a general algorithm and prove its possible convergence results.

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In Section 3, we focus on problems where the objective function and gradient are very expensive. For these cases, we advocate algorithms based on solving surrogate models at each iteration. Minimizing the surrogate models would be extremely expensive in scenarios where the objective function and gradient are cheap; however, in the presence of an expensive objective and gradient, the cost of the surrogate minimization becomes negligible. In this context, we propose an algorithm for minimizing sums of squares, where each component is approximated by a quadratic whose Hessian is updated at each iteration using quasi-Newton formulas, and the sum of squares of these approximations (a quartic function) is minimized at each main iteration. Finally, we introduce safeguards to fall within the general hypotheses of Section 2.

In Section 4, we apply the algorithms introduced in Section 3 to parameter estimation problems in natural channels.

Finally, in Section 5, we present our conclusions and outline directions for future research.

2 Optimization in the presence of forbidden sets

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Many optimization problems involve forbidden regions. This means parts of the domain where the objective function is not defined, is unreliable, or has no physical meaning. Sometimes these regions are excluded by introducing constraints to the problem, but on other occasions, resorting to constraints does not work. On the one hand, forbidden regions may not be well-characterized by constraints defined by algebraic equalities or inequalities. On the other hand, even when safe constraints can be identified, algorithms tend to hover near their infeasibility, eagerly searching for low values of the objective function, when the right move would be to abandon the forbidden region forcefully and, at the same time, sensibly. Other times, especially in the field of derivative-free optimization, algorithms include heuristic rules to avoid forbidden regions.

The algorithmic framework addressed in this paper is simple: we recognize that well-established algorithms exist for solving well-defined optimization problems, and we try to leverage their features as much as possible, instead of discarding them to look for radically different alternatives. The efficiency of this procedure lies in the modeler's ability to efficiently define the forbidden regions. Given that definition, the basic method consists of adopting a well-established algorithm—accepting that, occasionally, it may lead us to forbidden points—and treating those forbidden points as if they were points where the objective function is very large, but without using their value (which does not even need to be calculated or exist) for future computations. It is important to highlight that a strong motivation for employing forbidden region techniques is that objective functions can be very expensive to evaluate and, therefore, it is important to have criteria to avoid evaluating them when their evaluation promises to be useless.

Standard unconstrained minimization algorithms proceed, at each iteration, by calculating trial points that are accepted or rejected according to the value of the objective function. Even when rejected, these trial points contribute to the selection of subsequent trial points because they are frequently used in interpolation procedures that replace (simpler) backtracking steps. However, often before computing the first trial point in an iteration, the user possesses prior knowledge of the problem that indicates which regions are impossible or highly unlikely to contain a solution. Under these circumstances, and especially if the function evaluation is expensive, it can be advantageous to summarily reject unlikely trial points, avoiding useless searches on the boundary between the possible and the impossible.

This point of view is closely related to the classical notion of a trust region. Indeed, a trust region can be considered a region outside of which the occurrence of a solution is unlikely. If we possess a surrogate model of the objective function, it makes sense to minimize the model within the trust region to produce a trial point where evaluating the true function is worthwhile. However, an exhaustive search within the trust region to find a minimizer of the surrogate model can be expensive and inefficient. The results presented in this paper correspond to the case where the search within the trust region is reasonably exhaustive only if a solution promises to lie in the interior of the trust region, but not on its boundary.

Consider the problem

$$\text{Minimize } f(x) \text{ subject to } x \in \mathbb{R}^n \text{ and } x \notin P, \quad (1) \quad \{\text{problem}\}$$

where $P \subset \mathbb{R}^n$ is closed.

We assume that $\nabla f(x)$ exists and is continuous for all $x \in \mathbb{R}^n - P$. We denote $g(x) = \nabla f(x)$.

Algorithm 2.1.

Let $\theta \in (0, 1)$, $\alpha \in (0, 1/2)$, and $\beta > 0$ and $\delta > 0$ be algorithmic parameters. Let $x^0 \in \mathbb{R}^n - P$ be the initial approximation.

Step 0. Initialize $k \leftarrow 0$.

Step 1. If $\|g(x^k)\| = 0$, finish the execution of the algorithm.

Step 2. Compute $d^k \in \mathbb{R}^n$ such that

$$g(x^k)^T d^k \leq -\theta \|d^k\|_2 \|g(x^k)\|_2 \text{ and } \delta \geq \|d^k\| \geq \beta \|g(x^k)\|. \quad (2) \quad \{\text{conds}\}$$

Step 3. Define

$$t_{first,k} = 1 \quad (3) \quad \{\text{tuno}\}$$

or

$$t_{first,k} = \sup\{t \in [0, 1] \mid x^k + td^k \notin P\}. \quad (4) \quad \{\text{tsup}\}$$

Set $t \leftarrow t_{first,k}$.

Set $\nu = 1$.

Step 4. If

$$x^k + td^k \notin P \quad (5) \quad \{\text{notin}\}$$

and

$$f(x^k + td^k) \leq f(x^k) + \alpha tg(x^k)^T d^k \quad (6) \quad \{\text{armijo}\}$$

define $t_k = t$, $x^{k+1} = x^k + t_k d^k$, and $\nu_k = \nu$. Set $k \leftarrow k + 1$, and go to Step 1.

Step 5. Update $\nu \leftarrow \nu + 1$. Choose $t_{new} \in [0.1t, 0.9t]$, set $t \leftarrow t_{new}$, and go to Step 4.

Remark

Note that, under the choice (4), $x^k + t_{first,k} d^k$ may belong to P or not. If $t_{first,k} < 1$ we have that $x^k + t_{first,k} d^k \in P$, because, by the definition (4), there are elements of P of the form $x^k + td^k$ for t arbitrarily close to $t_{first,k}$ and P is closed. But, if $t_{first,k} = 1$, possibilities $(x^k + t_{first,k} d^k \in P$ and $x^k + t_{first,k} d^k \notin P)$ remain.

Lemma 2.1. *Every iteration of Algorithm 2.1 is well defined.*

Proof. By definition $x^0 \notin P$. Then, $g(x^k)$ exists and the algorithm stops at x^k if and only if $\|g(x^k)\| = 0$.

Assume that $x^k \notin P$ has been computed and $\|g(x^k)\| \neq 0$. Then, by (2),

$$g(x^k)^T d^k < 0. \quad (7) \quad \{\text{con}\}$$

We will prove that, after a finite number of inner steps, we can find $t > 0$ such that $x^k + td^k \notin P$ and (6) takes place.

Since $x^k \in \mathbb{R}^n - P$ and $\mathbb{R}^n - P$ is open, there exists $\delta > 0$ such that $x \in \mathbb{R}^n - P$ whenever $\|x - x^k\| \leq \delta$.

Then, there exists $\bar{t} > 0$ such that $\|(x^k + td^k) - x^k\| \leq \delta$ for all $t \in [0, \bar{t}]$. But, after a final number of reductions of t at Step 4 we have that $t \leq \bar{t}$, so the condition (5) takes place for all $t \leq \bar{t}$. Then, by (7), the definition of directional derivative and the continuity of the gradient, we have that

$$\lim_{t \rightarrow 0} \frac{f(x^k + td^k) - f(x^k)}{t} = g(x^k)^T d^k < 0$$

So, since $\alpha < 1$, for t small enough we have that

$$\frac{f(x^k + td^k) - f(x^k)}{tg(x^k)^T d^k} \geq \alpha.$$

Therefore, (6) holds for t small enough. So, after a finite number of reductions of t , the acceptance condition (6) takes place. This completes the proof. $\square$

Lemma 2.2 *Assume that for infinitely many indices $k \in K \subset \mathbb{N}$, $x^k + t_{first,k}d^k$ is chosen by (4) and belongs to P . Suppose that there exists ν_{max} such that $\nu_k \leq \nu_{max}$ for all $k \in K$ and $\lim_{k \rightarrow \infty} \|t_k d^k\| = 0$. Let x^* be a limit point of $\{x^k\}_{k \in K}$. Then $x^* \in P$.*

Proof. By the definition of the algorithm, $t_k \geq 0.1^{\nu_k} t_{first,k}$. Therefore, if $\|t_k d^k\|$ tends to zero and ν_k is bounded, we have that $\lim_{k \in K} \|t_{first,k} d^k\| = 0$. Now,

$$\|x^k + t_{first,k} d^k - x^*\| \leq \|x^k - x^*\| + \|t_{first,k} d^k\|,$$

So,

$$\lim_{k \in K} x^k + t_{first,k} d^k = x^*.$$

Since P is closed and $x^k + t_{first,k} d^k \in P$, this implies that $x^* \in P$. $\square$

Theorem 2.1. *If x^* is a limit point of the sequence generated by Algorithm 2.1, we have that $x^* \in P$ or $\|g(x^*)\| = 0$.*

Proof. let $K \subset \mathbb{N}$ be an infinite sequence of indices such that

$$\lim_{k \in K} x^k = x^*. \quad (8) \quad \{\text{limK}\}$$

If $x^* \in P$ there is nothing to prove. So, from now on we assume that

$$x^* \notin P \text{ and } \|g(x^*)\| > 0. \quad (9) \quad \{\text{absu}\}$$

We will prove that (9) is impossible.

Let us consider two cases:

Case 1: $\lim_{k \in K} \|t_k d^k\| = 0$.

Case 2: There exists a subsequence $K_1 \subset K$ and $c_1 > 0$ such that $\|t_k d^k\| > c_1$ for all $k \in K_1$.

Clearly, Case 2 is the negation of Case 1.

Assume that Case 1 takes place.

Suppose, for a moment, that for infinitely many iterations $k \in K_1 \subset K$, the initial trial step $t_{first,k}$ has been chosen according to (4) and, in addition, t_k was the accepted step at the first choice of t after the initial choice. In other words, $\nu_k = 2$ for all $k \in K_1$. Then, by Lemma 2.1, we have that $x^* \in P$. Therefore, this possibility can be discarded.

Therefore, for $k \in K$ large enough, the choice of t_k has been preceded by a choice of $t'_k \leq 10t_k$ such that (6) failed to hold when $t = t'_k$. Moreover, since $\lim_{k \in K} \|t_k d^k\| = 0$ we have that

$$\lim_{k \in K} \|t'_k d^k\| = 0. \quad (10) \quad \{\text{liteli}\}$$

So, for $k \in K$ large enough,

$$f(x^k + t'_k d^k) > f(x^k) + \alpha t'_k g(x^k)^T d^k \quad (11) \quad \{\text{noarm}\}$$

and

By (11) and the Mean Value Theorem, for all $k \in K$ large enough, there exists $\xi_k \in [0, 1]$ such that

$$g(x^k + \xi_k t'_k d^k)^T d^k > \alpha g(x^k)^T d^k. \quad (12) \quad \{\text{tvm}\}$$

By (10), dividing both members of (12) by $\|d^k\|$ and taking limits for $k \in K$ at a convergent subsequence of $d^k/\|d^k\|$ we get:

$$g(x^*)^T d \geq \alpha g(x^*)^T d \quad (13) \quad \{\text{thico}\}$$

where $\|d\| = 1$ and d is the limit of $d^k/\|d^k\|$ for a convergent subsequence. Since $0 < \alpha < 1$ This implies that

$$g(x^*)^T d \geq 0. \quad (14) \quad \{\text{thic}\}$$

But, by (2), we have that

$$g(x^k)^T d^k \leq -\theta \|g(x^k)\|_2 \|d^k\|_2,$$

So,

$$g(x^*)^T d \leq -\theta \|g(x^*)\|_2 < 0.$$

This contradicts (14). This contradiction comes from assuming that Case 1 takes place.

Therefore, we only need to analyze Case 2. By the definition of the algorithm we have that $x^k + t_k d^k \notin P$. Moreover, by (6),

$$f(x^k + t_k d^k) \leq f(x^k) + \alpha t_k g(x^k)^T d^k.$$

So, by (2),

$$f(x^k + t_k d^k) \leq f(x^k) - \alpha \theta t_k \|g(x^k)\|_2 \|d^k\|_2.$$

So, under the assumption that $\|g(x^*)\| > 0$, for $k \in K$ large enough we have that

$$f(x^k + t_k d^k) \leq f(x^k) - \alpha \theta c c_1. \quad (15) \quad \{\text{abajo}\}$$

Since $f(x^{k+1}) \leq f(x^k)$ for all $k \in \mathbb{N}$, (15) implies that $\lim f(x^k) = -\infty$, we contradicts the continuity of f at x^* . So, Case 2 is impossible too. Since both possibilities are impossible, it turns out that the assumption $\|g(x^*)\| > 0$ is impossible too. $\square$

3 Vamos ver se isto funciona:

$\{\text{vamosver}\}$

Nosso objetivo é minimizar uma função que é uma soma de quadrados:

$$\text{Minimizar } f(x) = \frac{1}{2} \sum_{i=1}^m f_i(x)^2 \quad (16) \quad \{\text{lasoma}\}$$

ou seja

$$\text{Minimizar } \frac{1}{2} \|F(x)\|_2^2,$$

com

$$F(x) = (f_1(x), \dots, f_m(x))^T.$$

Por Taylor, dado $a \in \mathbb{R}^n$, temos que

$$f_i(x) = f_i(a) + \nabla f_i(a)^T (x - a) + \frac{1}{2} (x - a)^T \nabla^2 f_i(a) (x - a) + O(\|x - a\|^3).$$

Portanto,

$$\begin{aligned} f_i(x)^2 &= [f_i(a) + \nabla f_i(a)^T (x - a) + \frac{1}{2} (x - a)^T \nabla^2 f_i(a) (x - a)]^2 \\ &+ 2[f_i(a) + \nabla f_i(a)^T (x - a) + \frac{1}{2} (x - a)^T \nabla^2 f_i(a) (x - a)] O(\|x - a\|^3) + O(\|x - a\|^6). \end{aligned}$$

Logo,

$$f(x) = \sum_{i=1}^m [f_i(a) + \nabla f_i(a)^T(x-a) + \frac{1}{2}(x-a)^T \nabla^2 f_i(a)(x-a)]^2 + f(a)O(\|x-a\|^3). \quad (17) \quad \{\text{apro}\}$$

Esta fórmula induz de ideia de minimizar, em cada iteração, a função $\sum_{i=1}^m [f_i(a) + \nabla f_i(a)^T(x-a) + \frac{1}{2}(x-a)^T \nabla^2 f_i(a)(x-a)]^2$. Mas isto é impraticável devido ao custo de $\nabla^2 f_i$.

Entretanto, podemos adotar a ideia de aproximar, via fórmulas quase-Newton, essas Hessianas. Isso leva à proposta de minimizar em cada iteração:

$$\text{Minimizar } \sum_{i=1}^m [f_i(a) + \nabla f_i(a)^T(x-a) + \frac{1}{2}(x-a)^T H_i(a)(x-a)]^2. \quad (18) \quad \{\text{thesu}\}$$

sujeita a uma região de confiança ou acrescentada de uma regularização.

(18) é a minimização de uma quártica, o que normalmente não é aconselhável. Mas no contexto de funções muito caras, esta ideia pode ser viável. Logo, para minimizar globalmente (18) não pouparíamos esforços.

Há muitas fórmulas quase-Newton. Por exemplo, as fórmulas BFGS e SR1, que são as seguintes:

$$H_+ = H + yy^T/s^T y - (Hss^T H)/(s^T Hs) \quad (19) \quad \{\text{bfgs}\}$$

e

$$H_+ = H + (y - Hs)(y - Hs)^T/(y - Hs)^T s. \quad (20) \quad \{\text{sr1}\}$$

Estas fórmulas são as que vão definir a direção d^k no Algoritmo 2.1, da seguinte maneira.

1. Inicializar $H_i = 0$ para todo $i = 1, \dots, m$.
2. Se $k = 0$, $d^k = -\nabla f(x^k)$.
3. Se $k > 0$ definir $s \leftarrow x^k - x^{k-1}$, Para cada $i = 1, \dots, m$ calcular $y = \nabla f_i(x^k) - \nabla f_i(x^{k-1})$. Para cada $i = 1, \dots, m$ calcular H_+ usando (19) ou (20). Definir $H \leftarrow H_+$. Resolver, globalmente, (18) com $a = x^k$, obtendo x . Definir $d^k = x - x^k$. Verificar as condições (2). Se não se verificam, substituir d^k pela direção de $-\nabla f(x^k)$. Espero que se verifiquem sempre, em caso contrário tanto trabalho foi jogado no lixo.

4 Complexity

{complexity}

Algorithm 4.1.

Let $x^0 \in \mathbb{R}^n - P$ be the initial approximation, $\sigma_{min} > 0$, $M > 0$, $0 < \theta < 1$.

Step 0. Initialize $k \leftarrow 0$.

Step 1. Set $\sigma \leftarrow 0$.

Step 2. Define, for all $i = 1, \dots, m$, $H_i \in \mathbb{R}^{n \times n}$ a symmetric matrix such that $\|H_i\| \leq M$.

Step 2. Solve, approximately,

$$\text{Minimize } \sum_{i=1}^m [f_i(x^k) + \nabla f_i(x^k)^T s + \frac{1}{2}s^T H_i s]^2 + \sigma \|s\|_2^2, \quad (21) \quad \{\text{regsub}\}$$

obtaining the trial step s^{trial} , that satisfies

$$\sum_{i=1}^m (f_i(x^k) + \nabla f_i(x^k)^T s^{trial} + \frac{1}{2}[s^{trial}]^T H_i s^{trial})^2 + \sigma \|s^{trial}\|_2^2 \leq f(x^k) \quad (22) \quad \{\text{bajomo}\}$$

and

$$\|\nabla_s \left(\sigma \|s^{trial}\|^2 + \sum_{i=1}^m (f_i(x^k) + \nabla f_i(x^k)^T s^{trial} + \frac{1}{2} [s^{trial}]^T H_i s^{trial})^2 \right)\| \leq \theta \|\nabla f(x^k)\| \quad (23) \quad \{\text{gcero}\}$$

Step 3. If

$$x^k + s^{trial} \notin P \quad (24) \quad \{\text{feasi}\}$$

and

$$f(x^k + s^{trial}) \leq f(x^k) - \alpha \|s^{trial}\|^2, \quad (25) \quad \{\text{armial}\}$$

define

$$s^k = s^{trial}, \quad (26) \quad \{\text{stsk}\}$$

$$x^{k+1} = x^k + s^k, \quad (27) \quad \{\text{novox}\}$$

set

$$k \leftarrow k + 1, \quad (28) \quad \{\text{next}\}$$

$$\sigma_k \leftarrow \sigma, \quad (29) \quad \{\text{sigmak}\}$$

and go to Step 1.

Otherwise, set

$$\sigma \leftarrow \max\{\sigma_{min}, 2\sigma\} \quad (30) \quad \{\text{upsig}\}$$

and go to Step 2.

Lemma 4.1 *If $\sigma > 0$ and $s \in \mathbb{R}^n$ is such that*

$$\sum_{i=1}^m (f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s)^2 + \sigma \|s\|_2^2 \leq f(x) \quad (31) \quad \{\text{bajom}\}$$

one has that

$$\|s\|_2^2 \leq f(x)/\sigma \quad (32) \quad \{\text{cotas}\}$$

Proof: By (31) and the fact that $\sum_{i=1}^m [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s]^2 \geq 0$ we have that $\sigma \|s\|^2 \leq f(x)$. $\square$

Assumption

There exist $\gamma > 0$ and $\delta_{lips} > 0$ such that, for all $x, z \notin P$ such that $\|x - z\| \leq \delta_{lips}$, and $i = 1, \dots, m$,

$$f_i(z) \leq f_i(x) + \nabla f_i(x)^T (z - x) + \frac{\gamma}{2} \|z - x\|^2. \quad (33) \quad \{\text{taylor}\}$$

Theorem 4.1. *Algorithm 4.1 is well defined.*

Proof: We need to prove that, for all x^k generated by Algorithm 4.1, the iterate x^{k+1} can be computed in finite time. Let us denote, for simplicity, $x = x^k$, $\|\cdot\| = \|\cdot\|_2$, and $H_i = H_{i,k}$.

By the definition of the Algorithm 4.1, $x^0 \in \mathbb{R}^n - P$ is well defined. Assume, as inductive hypothesis that $x^k \in \mathbb{R}^n - P$ is well defined. Since P is closed, there exists $\delta > 0$ such that $x^k + s \in \mathbb{R}^n - P$ whenever $\|s\| \leq \delta$. If the conditions (24) and (25) take place for some σ such that $\|s\| > \min\{\delta_{lips}, \delta\}$, the point $x^{k+1} = x^k + s$ is well defined and there is nothing to be proved. Therefore, we only need to consider the case in which $\|s^{trial}\| \leq \min\{\delta_{lips}, \delta\}$. For simplicity, let us denote $s = s^{trial}$.

By (33), for all $i = 1, \dots, m$ and $s \in \mathbb{R}^n$, we have that

$$f_i(x + s) \leq f_i(x) + \nabla f_i(x)^T s + \frac{\gamma}{2} \|s\|^2$$

Then, since $\|H_i\| \leq M$,

$$f_i(x+s) \leq f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s + M\|s\|^2 + \frac{\gamma}{2} \|s\|^2$$

So,

$$f_i(x+s)^2 \leq [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s]^2 + \left[M + \frac{\gamma}{2}\right]^2 \|s\|^4 + 2[f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s] \left[M + \frac{\gamma}{2}\right] \|s\|^2.$$

Therefore,

$$\sum_{i=1}^m f_i(x+s)^2 \leq m \left[M + \frac{\gamma}{2}\right]^2 \|s\|^4 + \sum_{i=1}^m [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s]^2 + 2[f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s] \left[M + \frac{\gamma}{2}\right] \|s\|^2.$$

Now, by (22), we have that

$$\sum_{i=1}^m [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s]^2 \leq \sum_{i=1}^m f_i(x)^2 - \sigma \|s\|^2,$$

then

$$\sum_{i=1}^m f_i(x+s)^2 \leq \sum_{i=1}^m f_i(x)^2 - \sigma \|s\|^2 + m \left[M + \frac{\gamma}{2}\right]^2 \|s\|^4 + 2 \left[M + \frac{\gamma}{2}\right] \|s\|^2 \sum_{i=1}^m [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s]$$

Therefore, $\sum_{i=1}^m f_i(x+s)^2 \leq \sum_{i=1}^m f_i(x)^2 - \alpha \|s\|^2$ whenever

$$-\sigma \|s\|^2 + m \left[M + \frac{\gamma}{2}\right]^2 \|s\|^4 + 2 \left[M + \frac{\gamma}{2}\right] \|s\|^2 \sum_{i=1}^m [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s] \leq -\alpha \|s\|^2.$$

This inequality holds whenever

$$m \left[M + \frac{\gamma}{2}\right]^2 \|s\|^4 + 2 \left[M + \frac{\gamma}{2}\right] \|s\|^2 \sum_{i=1}^m [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s] \leq (\sigma - \alpha) \|s\|^2.$$

A sufficient condition of the last inequality is:

$$\sigma \geq \alpha + m \left[M + \frac{\gamma}{2}\right]^2 \|s\|^2 + 2 \left[M + \frac{\gamma}{2}\right] \sum_{i=1}^m [f_i(x) + \nabla f_i(x)^T s + \frac{1}{2} s^T H_i s]$$

Since $\|s\| = \|s^{trial}\| \leq \min\{\delta_{lips}, \delta\}$, a sufficient condition for this inequality is:

$$\sigma \geq \alpha + m(M+1) \left[M + \frac{\gamma}{2}\right]^2 \min\{\delta_{lips}, \delta\}^2 + \left[\frac{2M+\gamma}{2}\right] \sum_{i=1}^m [|f_i(x)| + \|\nabla f_i(x)\| \min\{\delta_{lips}, \delta\}]. \quad (34) \quad \{\text{sigmar}\}$$

Clearly, this condition holds after a finite number of increases of σ at (30). $\square$

Assumption

There exists $c_g > 0$ such that for all $x \notin P$ and $f(x) \leq f(x^0)$ we have that

$$\|\nabla f_i(x)\| \leq c_g \quad (35) \quad \{\text{boundg}\}$$

for all $i = 1, \dots, m$.

Theorem 4.2. *There exists a constant $c_{eval} > 0$ that only depends of $\sigma_{min}, f(x^0), \alpha, m, M, \gamma$, and δ_{lips} such that, given any sequence generated by Algorithm 4.1, the number of of evaluations that is necessary to obtain the sufficient condition (25) at each iteration k is bounded above by c_{eval} .*

Proof: Assume that $x^k \in \mathbb{R}^n - P$ is an iterate generated by Algorithm 4.1. Then, as in the proof of Theorem 4.1, there exists $\delta > 0$ such that $x^k + s \in \mathbb{R}^n - P$ whenever $\|s\| \leq \delta$. By Lemma 4.1, after a finite number of increases of σ that do not involve evaluations of f , all the trial increments are such that $\|s^{trial}\| \leq \delta$. Therefore, function evaluations are associated only to the possible increase of the regularization parameters σ that produce increments whose size is, at most, δ . Now, by the rationale of Theorem 4.1 that leads to (34), if

$$\sigma \geq \alpha + m(M+1) \left[M + \frac{\gamma}{2} \right] \delta_{lips}^2 + \left[\frac{2M+\gamma}{2} \right] \left(\sum_{i=1}^m |f_i(x^k)| + mc_g \delta_{lips} \right), \quad (36) \quad \{\text{simen}\}$$

the condition (25) is fulfilled. Now,

$$\sum_{i=1}^m |f_i(x^k)| \leq \sqrt{\sum_{i=1}^m |f_i(x^k)|^2} \leq \sqrt{mf(x^0)}.$$

Then, (25) is fulfilled whenever

$$\sigma \geq c_{eval} \equiv \alpha + m(M+1) \left[M + \frac{\gamma}{2} \right] \delta_{lips}^2 + \left[\frac{2M+\gamma}{2} \right] (\sqrt{mf(x^0)} + mc_g \delta_{lips}). \quad (37) \quad \{\text{sime}\}$$

Therefore, the sufficient condition (25) is obtained increasing σ a number of times proportional to the logarithm of c_{eval} and σ_{min} . This completes the proof. $\square$

Remark. It must be warned that, in our case, the fact that the decrease can be obtained using a bounded number function evaluations do not imply that the regularization parameters σ_k are bounded. Indeed, we can say that, for obtaining $x^k + s^{trial}$ we only need a finite number of function evaluations, but we do not know how much the regularization parameter must be increased. This is a crucial difference between this complexity analysis and the one that is usual when forbidden points are not presented.

Theorem 4.3 *Let $\{x^k\}$ be a sequence generated by Algorithm 4.1. Then,*

$$\lim_{k \rightarrow \infty} \|s^k\| = 0. \quad (38) \quad \{\text{scero}\}$$

Moreover, given $\varepsilon > 0$, the number of iterations such that $\|s^k\| > \varepsilon$ is not bigger than $f(x^0)\varepsilon^{-2}/\alpha$.

Proof: We have already proved that, at each iteration of the algorithm, (25) holds. Therefore, for all $k \in \mathbb{N}$,

$$f(x^k) \leq f(x^0) - \alpha(\|s^0\|^2 + \dots + \|s^{k-1}\|^2). \quad (39) \quad \{\text{suma}\}$$

Since $f(x^k) \geq 0$ for all $k \in \mathbb{N}$, this implies (38).

Moreover, let N_k be the number indices such that $\|s^j\| > \varepsilon$ and $j \leq k$.

By (39) we have that, for all $k \in \mathbb{N}$,

$$f(x^k) \leq f(x^0) - \alpha N_k \varepsilon^2.$$

Since $f(x^k) \geq 0$, this implies that

$$\alpha N_k \varepsilon^2 \leq f(x^0)$$

for all $k \in \mathbb{N}$.

So, for all $k \in K$,

$$N_k \leq \frac{f(x^0)}{\alpha} \varepsilon^{-2}.$$

So, the desired result is proved. $\square$

Theorem 4.3 Assume that $\{x^k\}$ is a sequence generated by Algorithm 4.1. Then, the number of iterations at which $x^k + s \in \mathbb{R}^n - P$ for all $x \in B(x^k, \delta)$ and $\|\nabla f(x^k)\| \leq \varepsilon$ is bounded above by $\frac{f(x^0)}{\alpha} (\varepsilon/\bar{c})^{-2}$, where

$$\bar{c} = \frac{(2f(x^0)/\delta^2 + 2M\sqrt{m}\sqrt{f(x^0)} + 2mc_g^2 + 2mc_gM + Mmc_g + mM)}{1 - \theta}.$$

Proof: By (23), if $s = s^{trial}$ we have that

$$\begin{aligned} \theta \|\nabla f(x^k)\| &\geq \left\| \nabla_s \left(\sigma \|s\|^2 + \sum_{i=1}^m (f_i(x^k) + \nabla f_i(x^k)^T s + \frac{1}{2} s^T H_i s)^2 \right) \right\| \\ &= \left\| 2\sigma s + 2 \sum_{i=1}^m \left(f_i(x^k) + \nabla f_i(x^k)^T s + \frac{1}{2} s^T H_i s \right) (\nabla f_i(x^k) + H_i s) \right\| \end{aligned}$$

So, as $\nabla f(x^k) = 2 \sum_{i=1}^m f_i(x^k) \nabla f_i(x^k)$,

$$\begin{aligned} &\theta \|\nabla f(x^k)\| \\ &\geq \|\nabla f(x^k) + 2\sigma s + 2 \sum_{i=1}^m f_i(x^k) H_i s + 2 \sum_{i=1}^m (\nabla f_i(x^k)^T s) \nabla f_i(x^k) + 2 \sum_{i=1}^m (\nabla f_i(x^k)^T s) H_i s + \sum_{i=1}^m (s^T H_i s) \nabla f_i(x^k) + \sum_{i=1}^m (s^T H_i s) H_i s\|. \end{aligned}$$

So,

$$\begin{aligned} &(1 - \theta) \|\nabla f(x^k)\| \\ &\leq 2\sigma \|s\| + 2M \left| \sum_{i=1}^m f_i(x^k) \right| \|s\| + 2mc_g^2 \|s\| + 2mc_g M \|s\|^2 + Mmc_g \|s\|^2 + mM \|s\|^3 \\ &\leq 2\sigma \|s\| + 2M\sqrt{m}\sqrt{f(x^0)} \|s\| + 2mc_g^2 \|s\| + 2mc_g M \|s\|^2 + Mmc_g \|s\|^2 + mM \|s\|^3. \end{aligned}$$

Therefore,

$$\|\nabla f(x)\| \leq \frac{2\sigma \|s\| + 2M\sqrt{m}\sqrt{f(x^0)} \|s\| + 2mc_g^2 \|s\| + 2mc_g M \|s\|^2 + Mmc_g \|s\|^2 + mM \|s\|^3}{1 - \theta}.$$

Then, for all $k \in \mathbb{N}$,

$$\|\nabla f(x^k)\| \leq \frac{2\sigma \|s^{trial}\| + 2M\sqrt{m}\sqrt{f(x^0)} \|s^{trial}\| + 2mc_g^2 \|s^{trial}\| + 2mc_g M \|s^{trial}\|^2 + Mmc_g \|s^{trial}\|^2 + mM \|s^{trial}\|^3}{1 - \theta}.$$

We want to prove that σ_k is bounded independently of k . In our algorithm we start with $\sigma = 0$ and, if this σ fails in providing the required descent we try $\sigma = \sigma_{min}$. By (32) the maximum value of $\|s^{trial}\|$ at this point is bounded by $\sqrt{f(x)/\sigma_{min}}$. As far as we increase σ this bound decreases. After at most $\nu \equiv \log_2(f(x^0)/\sigma_{min})$ increases of σ we get $\|s^{trial}\| \leq 1$. Then, for all $\sigma \geq \sigma_{min} 2^\nu$, we have:

$$\|\nabla f(x^k)\| \leq \frac{(2\sigma + 2M\sqrt{m}\sqrt{f(x^0)} + 2mc_g^2 + 2mc_g M + Mmc_g + mM) \|s^{trial}\|}{1 - \theta}.$$

So,

$$\|\nabla f(x^k)\| \leq \frac{(2\sigma + 2M\sqrt{m}\sqrt{f(x^0)} + 2mc_g^2 + 2mc_g M + Mmc_g + mM) \|s^k\|}{1 - \theta}. \quad (40) \quad \{\text{nablame}\}$$

However we need to prove that σ_k is bounded. In the worst possible case, we need to increase σ up to the point at which $\|s^{trial}\| \leq \delta$.

For this purpose, by (32), a sufficient condition is that $f(x^0)/\sigma \leq \delta^2$. So, a sufficient condition for $\|s^k\| \leq \delta$ is that

$$\sigma \geq f(x^0)/\delta^2.$$

So, by (40),

$$\|\nabla f(x^k)\| \leq \frac{(2f(x^0)/\delta^2 + 2M\sqrt{m}\sqrt{f(x^0)} + 2mc_g^2 + 2mc_gM + Mmc_g + mM)\|s^k\|}{1 - \theta}.$$

Define

$$\bar{c} = \frac{(2f(x^0)/\delta^2 + 2M\sqrt{m}\sqrt{f(x^0)} + 2mc_g^2 + 2mc_gM + Mmc_g + mM)}{1 - \theta}.$$

Then,

$$\|\nabla f(x^k)\| \leq \bar{c}\|s^k\|.$$